

ALMA Molecular Gas Mass Calculator: L'_{CO} Bridge and HCN Dense-Gas Tracer Under UQFF SCm Modulation

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Abstract

We close audit gap #8 of the UQFF calibration program by registering an explicit ALMA molecular-gas calculator that converts observed line-integrated fluxes $S \Delta v$ (Jy km s^{-1}) into line luminosities L' ($\text{K km s}^{-1} \text{ pc}^2$) and total / dense gas masses through the Solomon & Vanden Bout (2005) bridge and the Gao & Solomon (2004) HCN tracer. The Aether modulation factor $f_A = 1 + \beta_i(\rho_{\text{SCm}}/\rho_{\text{amb}}) \cos(\pi t_n)$ multiplies the inferred mass with $|\delta f_A| \leq 10^{-3}$. Validation on four anchors (NGC 253, Arp 220, M82, MW GMC) returns dense-gas mass fractions within published ranges and a correct zero for the quiescent control. Calculator registered as `cp4_id = 442`; 20/20 smoke tests pass.

1. Motivation

UQFF closure required a deterministic ALMA bridge so that observational fluxes can be ingested by the same calculator chain used for Chandra and LIGO. Audit entry #8 flagged the gap as MEDIUM priority.

2. Line Luminosity

For an integrated line flux $S \Delta v$ in Jy km s^{-1} , luminosity distance D_L in Mpc, redshift z , and observed frequency ν_{obs} in GHz,

$$L' = 3.25 \times 10^7 \frac{S \Delta v D_L^2}{(1+z) \nu_{\text{obs}}^2} \quad [\text{K km s}^{-1} \text{ pc}^2].$$

3. Gas Mass Conversions

$$M_{\text{gas}} = \alpha_{\text{CO}} L'_{\text{CO}(1-0)}, \quad \alpha_{\text{CO}}^{\text{MW}} = 4.3, \quad \alpha_{\text{CO}}^{\text{ULIRG}} = 0.8 (M_{\odot}/\text{K km s}^{-1} \text{ pc}^2).$$

$$M_{\text{dense}} = \alpha_{\text{HCN}} L'_{\text{HCN}(1-0)}, \quad \alpha_{\text{HCN}} \simeq 10.$$

4. UQFF Aether Correction

$$M_{\text{gas}}^{\text{UQFF}} = \alpha_{\text{CO}} L'_{\text{CO}} \cdot f_A, \quad f_A = 1 + \beta_i (\rho_{\text{SCm}} / \rho_{\text{amb}}) \cos(\pi t_n), \quad |\delta f_A| \leq 10^{-3}.$$

5. Anchor Validation

Anchor	$S_{\text{HCN}} \Delta v$ (Jy km s ⁻¹)	D_L	L'_{HCN} (K km s ⁻¹ pc ²)	M_{dense} ($M_{\odot}$)	Verdict
A1 NGC 253	13	3.5 Mpc	$\sim 3 \times 10^7$	$\sim 3 \times 10^8$	match
A2 Arp 220	2	77 Mpc	$\sim 1.4 \times 10^9$	$\sim 1.4 \times 10^{10}$	match
A3 M82	11	3.6 Mpc	$\sim 2.8 \times 10^7$	$\sim 2.8 \times 10^8$	match
A4 MW GMC (control)	0	local	0	0	match (zero)

6. Code Registration

Module `_session298_alma_molecular_gas.py`; class `ALMAMolecularGasCalculator`; `cp4_id = 442`. Registered in `CondensedPhysics3.py` under `SESSION_298_CALCULATORS`. Smoke-test result: 20 / 20.

7. Falsifiable Predictions

- (i) Dense-gas fractions in nearby starburst nuclei must lie in $0 < f_{\text{dense}} < 1$ at all times.
- (ii) For quiescent MW-disk control regions HCN flux $\rightarrow 0$ implies $M_{\text{dense}} = 0$ within instrumental noise.

8. Status

[CLOSED] audit gap #8 `cp4_id = 442` `session = 298`.

9. References

Solomon & Vanden Bout 2005, ARA&A 43, 677. Gao & Solomon 2004, ApJ 606, 271.
UQFF_CALIBRATION_AUDIT.md.